

RCA ID	Root Cause	Error Code	Error Message	Resolution
RCA-101	Checkout transactions started failing after a payment gateway security update. The Stripe webhook signing secret configured in the application was outdated and no longer matched the active secret configured in the Stripe dashboard. As a result, webhook signature verification failed during payment confirmation, preventing successful order completion.	PAY-4017	Webhook signature verification failed. No matching signing secret found for request.	Updated the latest Stripe webhook signing secret in AWS Secrets Manager, synchronized the secret across production instances, restarted payment services, validated using Stripe CLI, and implemented automated secret rotation monitoring.
RCA-102	Redis cache nodes experienced memory exhaustion due to unlimited cache growth because cache keys were created without TTL.	REDIS-1009	OOM command not allowed when used memory > maxmemory.	Configured volatile-lru policy, enforced TTLs, cleaned stale keys, optimized cache generation, and added monitoring.
RCA-103	Celery workers became blocked by PostgreSQL deadlocks, causing large task queues.	DB-4003	ERROR: deadlock detected while waiting for ShareLock on transaction.	Implemented pagination, reduced transaction scope, added retries with exponential backoff, optimized indexes, and tuned Celery concurrency.
RCA-104	AWS S3 uploads failed after incorrect IAM policy changes removed PutObject permission.	S3-4031	AccessDenied: User is not authorized to perform s3:PutObject.	Rolled back IAM policies, validated permissions, enabled CloudTrail auditing, and added deployment validation.
RCA-105	Database connections leaked because exception paths did not close pooled connections.	DB-5002	Connection pool exhausted. Timeout waiting for idle connection.	Wrapped database operations in try-finally, enabled leak detection, tuned pool size, and added health monitoring.
RCA-106	Node.js image processing library leaked native memory causing container crashes.	NODE-6008	FATAL ERROR: JavaScript heap out of memory.	Updated dependency, rebuilt containers, enabled heap profiling, and tuned Kubernetes memory settings.
RCA-107	Disk usage reached 100% because search indexes and logs were never cleaned.	FS-7005	No space left on device.	Configured log rotation, scheduled cleanup jobs, added storage monitoring, and capacity planning.
RCA-108	Twilio trial account expired causing SMS verification failures.	SMS-4020	Twilio Error 20003: Authentication failed or account inactive.	Enabled auto-renew billing, upgraded account, configured alerts, and added SMS failover.

RCA-109	JWT serialization mismatch across services caused authentication failures.	AUTH-4015	JWT validation failed. Unsupported serialization format detected.	Standardized JWT libraries and algorithms, regenerated certificates, added compatibility tests.
RCA-110	Kafka producer batch configuration caused excessive consumer lag.	KAFKA-3006	Record batch size exceeded optimal threshold. Consumer lag increasing.	Optimized producer batch size, enabled Snappy compression, tuned linger.ms, and rebalanced consumers.